La subiectul de substitutie (3p din total):
Incadrarea lui T(n) intr-o clasa de complexitate: 1.5p
Demonstrarea, prin metoda substitutiei, a complexitatii lui T(n): 1.5p

Rezolvare substitutie 1:
T(n) = T(n/4) + T(n/3) + n => T(n) ∈ Theta(n) -> arbore de recurenta
$T(n / 4^{h1}) = T(1) \Rightarrow 4^{h1} = n \Rightarrow h_1 = \log_4(n)$
$T(n / 3^{h2}) = T(1) \Rightarrow 3^{h2} = n \Rightarrow h_2 = \log_3(n)$
Insumarea termenilor pe nivelul k ($k < h_1 < h_2$) va da: $(7^i * n) / 12^i$

$$\sum_{i=0}^{h1} \frac{7^i * n}{12^i} \leq T(n) \leq \sum_{i=0}^{h2} \frac{7^i * n}{12^i}$$

$$\sum_{i=0}^{log_4(n)} \frac{7^i * n}{12^i} \leq T(n) \leq \sum_{i=0}^{log_3(n)} \frac{7^i * n}{12^i}$$

$$n * \sum_{i=0}^{log_4(n)} \left(\frac{7}{12}\right)^i \leq T(n) \leq n * \sum_{i=0}^{log_3(n)} \left(\frac{7}{12}\right)^i$$

$n * (1 - (7/12)^{log4(n) + 1}) / (1 - 7/12) \leq T(n) \leq n * (1 - (7/12)^{log3(n) + 1}) / (1 - 7/12)$
$12n / 5 - 7n / 5 * n^{log4(7/12)} \leq T(n) \leq 12n / 5 - 7n / 5 * n^{log3(7/12)}$
Rezulta ca T(n) ∈ Θ(n)

Trebuie demonstrat:
c1 * n <= T(n) <= c2 * n

Cazul de baza:
c1 * 1 <= 1 <= c2 * 1
Deci, c1 <= 1 si c2 >= 1

Pas de inductie:
Presupun T(n/3) "A" (evident ca si T(n/4) "A"):
c1 * n / 3 <= T(n / 3) <= c2 * n / 3
c1 * n / 4 <= T(n / 4) <= c2 * n / 4
---------- (+)
c1 * (7n / 12) <= T(n / 3) + T(n / 4) <= c2 * (7n / 12) | + n
c1 * (7n / 12) + n <= T(n / 3) + T(n / 4) + n <= c2 * (7n / 12) + n
c1 * (7n / 12) + n <= T(n) <= c2 * (7n / 12) + n
c1 * n <= c1 * (7n / 12) + n <= T(n) <= c2 * (7n / 12) + n <= c2 * n

I. c1 * n <= c1 * (7n / 12) + n
   c1 * n <= n (7c1 / 12 + 1) | :n
   c1 <= 7c1 / 12 + 1
   5c1 / 12 <= 1
   c1 <= 12 /5 = 2.4
   Cum din cazul de baza c1 <= 1, rezulta ca c1 ∈ (0; 1]

II. c2 * (7n / 12) + n <= c2 * n
n * (7c2 / 12 + 1) <= c2 * n | :n
7c2 / 12 + 1 <= c2
1 <= 5c2 / 12
c2 >= 12 / 5 = 2.4
Cum din cazul de baza c2 >= 1, rezulta ca c2 ∈ [2.4; ∞)

Din I si II => T(n) ∈ Θ(n)

Rezolvare substitutie 2:
T(n) = T(2n/3) + T(n/3) + n^2 => T(n) ∈ Θ(n^2) -> arbore de recurenta
T(n / 3^{h1}) = T(1) => 3^{h1} = n => $h_1 = \log_3(n)$
T(2^{h2} * n / 3^{h2}) = T(1) => $(3/2)^{h2}$ = n => $h_2 = \log_{3/2}(n)$
Insumarea termenilor pe nivelul k ($k < h_1 < h_2$) va da: ($5^i * n^2$) / 9^i

$$\sum_{i=0}^{h1} \frac{5^i * n^2}{9^i} \leq T(n) \leq \sum_{i=0}^{h2} \frac{5^i * n^2}{9^i}$$

$$\sum_{i=0}^{log_3(n)} \frac{5^i * n^2}{9^i} \leq T(n) \leq \sum_{i=0}^{log_{3/2}(n)} \frac{5^i * n^2}{9^i}$$

$$n^2 * \sum_{i=0}^{log_3(n)} \left(\frac{5}{9}\right)^i \leq T(n) \leq n^2 * \sum_{i=0}^{log_{3/2}(n)} \left(\frac{5}{9}\right)^i$$

$n^2 * (1 - (5/9)^{log3(n) + 1}) / (1 - 5/9) \leq T(n) \leq n * (1 - (5/9)^{log3/2(n) + 1}) / (1 - 5/9)$
$9n / 4 - 9n / 4 * n^{log3(5/9)} \leq T(n) \leq 9n / 4 - 9n / 4 * n^{log3/2(5/9)}$
Rezulta ca T(n) ∈ Θ(n^2)

Trebuie demonstrat:
c1 * n^2 <= T(n) <= c2 * n^2

Cazul de baza:
c1 * 1 <= 1 <= c2 * 1
Deci, c1 <= 1 si c2 >= 1

Pas de inductie:
Presupun T(2n / 3) “A” (evident ca si T(n / 3) “A”)
c1 * (2n / 3)^2 <= T(2n / 3) <= c2 * (2n / 3)^2
c1 * (n / 3)^2 <= T(n / 3) <= c2 * (n / 3)^2
---------- (+)
c1 * (5n^2 / 9) <= T(2n / 3) + T(n / 3) <= c2 * (5n^2 / 9) | + n^2
c1 * (5n^2 / 9) + n^2 <= T(2n / 3) + T(n / 3) + n^2 <= c2 * (5n^2 / 9) + n^2
c1 * (5n^2 / 9) + n^2 <= T(n) <= c2 * (5n^2 / 9) + n^2
c1 * n^2 <= c1 * (5n^2 / 9) + n^2 <= T(n) <= c2 * (5n^2 / 9) + n^2 <= c2 * n^2

I. $c1 * n^2 <= c1 * (5n^2 / 9) + n^2$
$c1 * n^2 <= n^2 * (5c1 / 9 + 1) \mid : n^2$
$c1 <= 5c1 / 9 + 1$
$4c1 / 9 <= 1$
$c1 <= 9 / 4 = 2.25$
Cum din cazul de baza $c1 <= 1$, rezulta ca $c1 \in (0; 1]$

II. $c2 * (5n^2 / 9) + n^2 <= c2 * n^2$
$n^2 * (5c2 / 9 + 1) <= c2 * n^2 \mid : n^2$
$5c2 / 9 + 1 <= c2$
$1 <= 4c2 / 9$
$c2 >= 9 / 4 = 2.25$
Cum din cazul de baza $c2 >= 1$, rezulta ca $c2 \in [2.25; \infty)$

Din I si II => $T(n) \in \Theta(n^2)$

La subiectul de Master (2p din total):
Identificare a, b si $n^{(\log_b a)}$: 0.4p
Testarea cazului potrivit de Master: 1.2p
Aplicarea cazului potrivit de Master: 0.4p

Rezolvare Master 1:
$T(n) = 2T(n/2) + n*\log\log n + n*\log n$

$a = 2, b = 2, f(n) = n * \log(\log(n)) + n * \log(n)$
$n^{(\log_b a)} = n^{(\log_2 2)} = n$

Testam cazul 2 de Master:
$f(n) \in \Theta(n^{(\log_b a)} * \log^k n)$, adica $f(n) \in \Theta(n * \log^k n)$
$\lim_{n \to \infty} f(n) / (n * \log^k n) = \lim_{n \to \infty} (n * \log(\log(n)) + n * \log(n)) / (n * \log^k n) =$
$= \lim_{n \to \infty} (\log(\log(n)) + \log(n)) / \log^k n = \lim_{n \to \infty} \log(\log(n)) / \log^k n + \lim_{n \to \infty} \log(n) / \log^k n =$
= (Aplic L'Hopital pentru ambii termeni) $\lim_{n \to \infty} (1 / n * \log n) / (k * \log^{k-1} n * 1 / n) + \lim_{n \to \infty}$
$(1 / n) / (k * \log^{k-1} n * 1 / n) = \lim_{n \to \infty} 1 / (k * \log^k n) + \lim_{n \to \infty} 1 / (k * \log^{k-1} n)$
Pentru k = 0 => $\lim_{n \to \infty} 1 / (0 * \log^0 n) + \lim_{n \to \infty} 1 / (0 * \log^{-1} n)$ impartire la 0 -> ∄
Pentru k = 1 => $\lim_{n \to \infty} 1 / (1 * \log^1 n) + \lim_{n \to \infty} 1 / (1 * \log^0 n) = 0 + 1 = 1$ ✓

Pentru k = 1, $f(n) \in \Theta(n * \log^k n)$, adica $f(n) \in \Theta(n * \log n)$, deci se poate aplica al 2-lea caz de Master: $T(n) \in \Theta(n * \log^{k+1} n) \Rightarrow T(n) \in \Theta(n * \log^2 n)$

Rezolvare Master 2:
$T(n) = 3T(2n/3) + 3n/2*\log n$

$a = 3, b = 3/2, f(n) = (3n/2) * \log(n)$
$n^{(\log_b a)} = n^{(\log_{3/2} 3)}$

Testam cazul 1 de Master:
$f(n) \in O(n^{(\log_{3/2} 3 - \varepsilon)})$
$\lim_{n \to \infty} f(n) / (n^{(\log_{3/2} 3 - \varepsilon)}) = \lim_{n \to \infty} ((3n/2) * \log(n)) / (n^{(\log_{3/2} 3 - \varepsilon)}) = \lim_{n \to \infty} (3/2 * \log(n)) / (n^{(\log_{3/2} 2 - \varepsilon)}) =$ (Aplic L'Hopital) $\lim_{n \to \infty} (3/2) / ((\log_{3/2} 2 - \varepsilon) * \ln 2 * n^{(\log_{3/2} 2 - \varepsilon)}) = 0$, daca $\varepsilon \in (-\infty; \log_{3/2} 2)$
Cum $\varepsilon > 0 \Rightarrow \varepsilon \in (0; \log_{3/2} 2)$
Pentru $\varepsilon \in (0; \log_{3/2} 2)$, $f(n) \in O(n^{(\log_{3/2} 3 - \varepsilon)})$, deci pot aplica primul caz de Master:
$T(n) \in \Theta(n^{(\log_b a)}) \Rightarrow T(n) \in \Theta(n^{(\log_{3/2} 3)})$

La subiectul de Decidabilitate (2.5p din total):
Mentionarea clasei si argumentare: 0.5p
Demonstratie prin reducerea Turing: 1p
Demonstratie pentru incadrarea problemei in RE: 1p

Rezolvare Decidabilitate 1:
f(P) = 1, E x ai p(x) si p(x+1) se opresc

f(P) = {
1, $p(x) \neq \perp$ si $p(x+1) \neq \perp$
0, altfel
}
De demonstrat: $f_h <=_T f$
Pentru fiecare intrare (P, w) trebuie sa gasim P' a.i $P'(x) \neq \perp$ si $P'(x+1) \neq \perp$
$f_h(P, w) = 1 \Rightarrow P(w) \neq \perp$
$f(P') = 1 \Leftrightarrow P'(x) \neq \perp$ si $P'(x+1) \neq \perp$

Construim programul P':
```
P'(y) {
   P(w);
   return 1;
```

}
“=>” $f_h(P, w) = 1 \Rightarrow P(w) = 1 \Rightarrow P'(y) \neq \perp \Rightarrow P'(y) = 1, \forall y \Rightarrow \exists x$ a.i. $p'(x) \neq \perp$ si $p'(x+1) \neq \perp \Rightarrow f(P') = 1$
“<=” $f(P') = 1 \Rightarrow \exists x$ a.i. $p'(x) \neq \perp$ si $p'(x+1) \neq \perp \Rightarrow P(w) \neq \perp \Rightarrow f_h(P, w) = 1$
$f_h(P, w) <=_T f(P) \Rightarrow f \notin R$

Alternativa P’:

```
P’(y) {
        if x == 0 || x == 1
                P(w);
                return 1;
        else
                loop;
```

Fie programul:

```
Pf(p) {
   for (t = 1..∞) {
     for (x = 0..t) {
       if (p(x) ≠ ⊥ in t pasi sau mai putin && p(x+1) ≠ ⊥ in t pasi sau mai putin) {
          return 1;
       }
     }
   }
   return 0;
}
```

Asadar, $f \in RE \setminus R$

Rezolvare Decidabilitate 2:

```
f(p) = {
           1, ∀x ∈ ℵ, x <= 2021 , p(x) = x!
           0, altfel
         }
```

De demonstrat: $f_h <=_T f$
Pentru fiecare intrare (p, x) trebuie sa gasim p’ a.i $p(x) \neq \perp \Leftrightarrow f(p') = 1$
Construim programul p’:

```
p’(y) {
   p(x);
   result = 1;
   for i = 1..y:
     result *= i;
```

```
    return result;
}
```
“=>” $f_h(p, x) = 1 \Rightarrow p(x) \neq \perp \Rightarrow p'(y) = y!, \forall y \Rightarrow f(p') = 1$
“<=” $f(p') = 1 \Rightarrow \forall y <= 2021\ p'(y) = y! \Rightarrow p(x) \neq \perp \Rightarrow f_h(p, x) = 1$
$f_h <=_T f \Rightarrow f \notin R$

Fie programul:

```
P_f(p) {
    for (x = 0..2021) {
      if (p(x) ≠ x!) {
         while (true);
      }
    }
    return 1;
}
```
Asadar, $f \in RE \setminus R$

La subiectul de Amortizata (metoda creditelor) (2.5p din total):
Alegerea $\hat{C}_i$ pentru metoda creditelor: 0.5p (0.25p pt fiecare)
Verificare pentru cazurile de la inceput: 0.5p
Calcularea sumelor cu cost: 1p (cate 0.5p pentru fiecare)
Verificare inegalitate si concluzie: 0.5p

La subiectul de Amortizata (metoda analizei agregate) (2.5p din total):
Calculare T(n): 2p (0.75p impartire sume, 1.25p calculare sume)
Calculare $\hat{C}_i$ si concluzie: 0.5p

Rezolvare Amortizata 1:

```
C_i =  {
        sqrt(i), i=k^2x`
        1, altfel
        }
```

Metoda creditelor:

	pt pf	!(pt pf)
C_i	$\sqrt{i}$	1
$\hat{C}_i$	1	2

i = 1 => C_i = 1; $\hat{C}_i$ = 1; sum_diff = 0
i = 2 => C_i = 1, $\hat{C}_i$ = 2; sum_diff = 1
i = 3 => C_i = 1; $\hat{C}_i$ = 2; sum_diff = 2
i = 4 => C_i = 2; $\hat{C}_i$ = 1; sum_diff = 1

$$\sum_{i=1}^{n} \hat{C}_i \geq \sum_{i=1}^{n} C_i, \forall n$$

$$\sum_{i=1}^{n} \hat{C}_i = \sum_{i=1}^{\sqrt{n}} \hat{C}_{ptpf} + \sum_{i=1}^{n-\sqrt{n}} \hat{C}_{!ptpf} = \sum_{i=1}^{\sqrt{n}} 1 + \sum_{i=1}^{n-\sqrt{n}} 2 = 2n - \sqrt{n}$$

$$\sum_{i=1}^{n} C_i = \sum_{i=1}^{\sqrt{n}} \hat{C}_{ptpf} + \sum_{i=1}^{n-\sqrt{n}} \hat{C}_{!ptpf} = \sum_{i=1}^{\sqrt{n}} i + \sum_{i=1}^{n-\sqrt{n}} 1 = n/2 + \sqrt{n} / 2 + n - \sqrt{n} = 3n/2 - \sqrt{n} / 2$$

$2n - \sqrt{n} \geq 3n / 2 - \sqrt{n} / 2 \Leftrightarrow n/2 \geq \sqrt{n} / 2 \Leftrightarrow n \geq \sqrt{n}$, “A” $\forall$ n
Rezulta ca $\hat{C}_i = 2 \in \Theta(1)$

Analiza agregata:

$$T(n) = \sum_{i=1}^{n} C_i = \sum_{i=1}^{\sqrt{n}} \hat{C}_{ptpf} + \sum_{i=1}^{n-\sqrt{n}} \hat{C}_{!ptpf} = \sum_{i=1}^{\sqrt{n}} i + \sum_{i=1}^{n-\sqrt{n}} 1 = n/2 + \sqrt{n} / 2 + n - \sqrt{n} =$$

$3n/2 - \sqrt{n} / 2 \Rightarrow \hat{C}_i = T(n) / n \Rightarrow \hat{C}_i = 3/2 - 1/(2\sqrt{n}) \Rightarrow \hat{C}_i \simeq 3/2 \in \Theta(1)$

Rezolvare Amortizata 2:
C_i = {
i, i = k!
1, altfel
}

Metoda creditelor:

	fact	!(fact)
C_i	i	1
$\hat{C}_i$	1	3

i = 1 => C_i = 1; $\hat{C}_i$ = 1; sum_diff = 0
i = 2 => C_i = 2, $\hat{C}_i$ = 1; sum_diff = -1
i = 3 => C_i = 1; $\hat{C}_i$ = 3; sum_diff = 1
i = 4 => C_i = 1; $\hat{C}_i$ = 3; sum_diff = 3

i = 5 => C_i = 1; $\hat{C}_i$ = 3; sum_diff = 5
i = 6 => C_i = 6; $\hat{C}_i$ = 1; sum_diff = 0

$$\sum_{i=1}^{n} \hat{C}_i \geq \sum_{i=1}^{n} C_i, \forall n$$

$$\sum_{i=1}^{n} \hat{C}_i = \sum_{i=1}^{k} \hat{C}_{fact} + \sum_{i=1}^{n-k} \hat{C}_{!fact} = \sum_{i=1}^{k} 1 + \sum_{i=1}^{n-k} 3 = 3n - 2k, \text{ unde } k! = n, \text{ deci } k << n$$

$$\sum_{i=1}^{n} C_i = \sum_{i=1}^{k} \hat{C}_{fact} + \sum_{i=1}^{n-k} \hat{C}_{!fact} = \sum_{i=1}^{k} i! + \sum_{i=1}^{n-k} 1 = \sum_{i=1}^{k} i! + n - k$$

Fie secventa de operatii de la k!+1 la (k+1)!

$$\sum_{i=k!+1}^{(k+1)!} \hat{C}_i - \sum_{i=k!+1}^{(k+1)!} C_i \geq 0 \text{ “A” => } \sum_{i=k!+1}^{(k+1)!} \hat{C}_i \geq \sum_{i=k!+1}^{(k+1)!} C_i$$

Pot sa iau $\hat{C}_i$ pentru factorial = 0 (usureaza demonstratia)
3 * ((k+1)! - k! - 1) >= (k+1)! + (k+1)! - k! - 1
3k * k! - 3 >= (k+1)! + k * k! - 1
3k * k! >= (k+1)! + k * k! + 2
2k * k! >= (k+1)! + 2
2k * k! - (k+1) * k! >= 2
k! * (k-1) >= 2, $\forall$ k >= 2
Cum k! = n, => $\sum_{i=1}^{n} \hat{C}_i \geq \sum_{i=1}^{n} C_i$, $\forall$ n >= 2
Rezulta ca $\hat{C}_i \cong 3 \in \Theta(1)$

Analiza agregata:

$$T(n) = \sum_{i=1}^{n} C_i = \sum_{i=1}^{k} \hat{C}_{fact} + \sum_{i=1}^{n-k} \hat{C}_{!fact} = \sum_{i=1}^{k} i! + \sum_{i=1}^{n-k} 1 = \sum_{i=1}^{k} i! + n - k$$

$$\leq (k+1)! + n - k = (k+1)*n + n - k = (k+2) * n - k$$

$\hat{C}_i$ = T(n) / n => $\hat{C}_i$ = k + 2
Cum k << n => $\hat{C}_i \in \Theta(1)$